

Artificial Neural Network in FPGA for Temperature Prediction

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Abstract. In this work a temperature predictor has been designed. The prediction is made by an artificial neural network multilayer perceptron. Initially, the floating point algorithm was evaluated. Afterwards, the fixed point algorithm was designed on a Field Programmable Gate Array (FPGA). The architecture was fully parallelized and a maximum delay of 74 ns was obtained. The design tool used is System Generator of Xilinx.

Keywords: neural network, FPGA, temperature prediction, Simulink, System Generator, floating point, fixed point, VHDL, Verilog.

1 Introduction

Weather forecasting systems have been vastly studied since the beginning of the human being, for its relevance as predictors of important weather changes. As a science, Aristotle wrote “Meteorological” about 340 B.C. and so he is considered the founder of meteorology. Nowadays, weather forecasting have become a common tool in our daily routing as basic information for the day to come and for the next days. In the field of energy production techniques dependent on meteorological conditions such as sun or wind, meteorological systems provide the information necessary for compute the expected amount of produced energy. For example, insolation prediction methods are used on different systems. Numerous authors such as [1], [2], and [3] used neural networks in combination with fuzzy logic techniques for weather forecasting purposes. As insolation is strongly influenced by the monthly distribution of atmospheric pressure and temperature [4], they implemented in 2007 a simple predicting system based on neural networks using this information from the last 16 days as input, achieving error rates of around 15% in solar irradiation levels of the next 24 hours. An example of the effects of different meteorological conditions on the production of energy by a solar plant was presented in [5].

Being able to predict the occurrence of any event sufficiently in advance, can anticipate actions that produce results with lower costs. In [6], this paper describes the use of Artificial Neural Networks (ANN) and the prediction of time series modeling for the prediction of climatic variables.

In recent years Artificial Neural Networks (ANN) have been successfully applied as a tool in the prediction of time series, mainly due to the ability of ANN to capture the linear and nonlinear relationships between the data [7].

A temporal serie is a consecutive set of values that represent the dynamics of a phenomenon [8], $S = \{x_1, x_2, x_3, \dots, x_n\}$. One of the main objectives of the study is time series prediction, which is to estimate a value x_{p+1} from the previous p values. The time series forecasting is based on the assumption that given a discrete set of data with respect to time and for the same phenomenon, future values are dependent on past values [9], so the challenge is to determine what are these relations of dependency.

The prediction from time series has many applications in different areas of science, finance, trade and industry. For example, predicting the demand for a product is useful for optimizing the supply chain. Similarly, the forecast of electricity demand can save operating costs in generating stations [10]. In the present case, the time series correspond to meteorological data with which you want to predict future values of some climatic variables.

In this work, authors present an approach for the temperature prediction based on real signals from meteorological station. The innovation of this work has been the implementation of a fixed point system, programmed on a Field Programmable Gate Array (FPGA), in order to do a no-linear prediction based on Artificial Neural Network.

2 Floating Point Neural Network Design

The data used in this paper come from weather stations located in different sites of Costa Rica, and includes information since the 2005 with a sampling frequency of every half hour. In relation to the data, it is important to note that Costa Rica is characterized by a region with different climatic conditions (microclimate) and much variability.

Each specific place requires its own prediction model. The factors involved in the design will depend on the variability of the place. For example, if the temperature is relatively stable, prediction could be made only with the series of temperature data, but if it is highly variable, then it needed other data such as humidity, solar radiation, time, day, month, wind, among others.

For the construction of the ANN, the cases were obtained using a sliding window on each data series. The data enclosed by the window represent a study case, on a recurring basis, the window will move one to one on the data series until to obtain all the patterns, see Fig. 1.

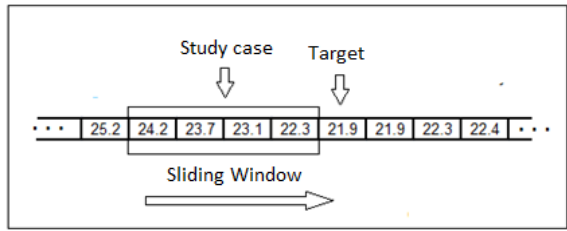


Fig. 1. Getting a study case from a time series

For the model presented here, the database corresponds to the weather station of the University of Costa Rica, located in the city of Turrialba. From the measurements recorded by the meteorological station and through the sliding window technique, we constructed a database consisting of 87,600 cases (five years).

The base of the model is an Artificial Neural Network Multilayer Perceptron [11] (ANN-MLP) Feed-Forward with Back-Propagation training algorithm with only one hidden layer (eight neurons). The Artificial Neural Network has five inputs, and the output layer has only one element that indicates the estimated value of temperature, see Fig. 2. To train the ANN we used data from a sampling year, which corresponds to 20% of the database.



Fig. 2. Multilayer Perceptron Neural network structure used

To determine the optimal number of neurons in the hidden layer is followed a trial and error process, which combines the number of neurons and the maximum allowable error. The number of neurons in the hidden layer is an important parameter for the effectiveness of the model [12], because if the number is small, the network does not properly reflect the relationships between data. Moreover, if the number of neurons is high, it produces an overfitting and the ANN cannot be general enough to be effective and tends to fail with new cases. The system was trained with the Neural Network Toolbox of Matlab [13]. The unipolar sigmoid activation function was specified for all neurons. At the end of training, synaptic weights and bias were obtained for all neurons in floating point format.

After designing the ANN, the results of the predictions were monitored and contrasted with true values. Fig. 3 shows graphs of true and estimated temperature values, the error is less than one degree Celsius.

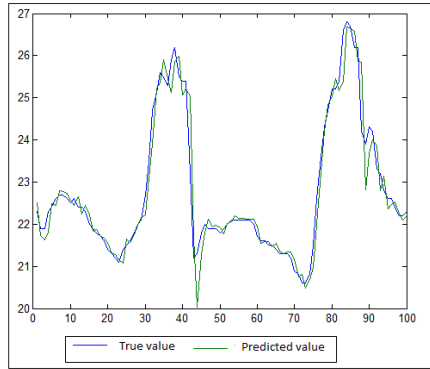


Fig. 3. True temperature value and estimated value by the neural network model

3 Neural Network Design on FPGA

The algorithm should be developed in a digital circuit using fixed point arithmetic in two's complement. It was designed on a FPGA [14]. The tool used is System Generator [15] of Xilinx manufacturer, which allows design in the form of block diagrams in a fast and flexible way.

FPGA devices are optimal for prototyping design because they have reasonable cost and good physical performances. The design method chosen should minimize the time spent by the designer, on the other side should be a flexible method. Xilinx is one of the most extended manufacturer of FPGA. Xilinx offers System Generator that is a design tool on Simulink [16] of Matlab. This tool allows the fast design of systems using block diagrams. The compilation generates the files necessary for the Integrated System Environment (ISE) [17] of Xilinx for FPGA, where the description of the circuit is obtained in a standard Hardware Description Language (HDL). These languages are Verilog [18] and VHDL (Very High Speed Integrated Circuit Hardware Description Language) [19].

The system designed is a Time Delay Neural Network (TDNN), it was developed in three phases. Firstly its architecture is specified, afterwards the values and parameters are initialized; finally, the TDNN functionality is evaluated. The TDNN is shown in Fig. 4. The input data is an unsigned 9 bits with 4 decimals for covering the range and the resolution of temperature with a maximum error of 1%. There are five input buses to the Neural Network in the FPGA. The connectivity between the layers is done with Simulink Signal Routing blocks, for this reason the design time is minimized. The intermediate layer is composed of eight neurons, each neuron has five inputs. In the first stage of these neurons the inputs are multiplied by synaptic weights and the bias is added with a combinational circuit. This operation is fully parallelized. In a second stage the activation function is designed using a Read Only Memory (ROM) where 128 values of unipolar sigmoid are stored for having enough resolution. The last layer is composed of one neuron with eight inputs, its activation function is the identity function. The design is completely combinational so does not require any clock.

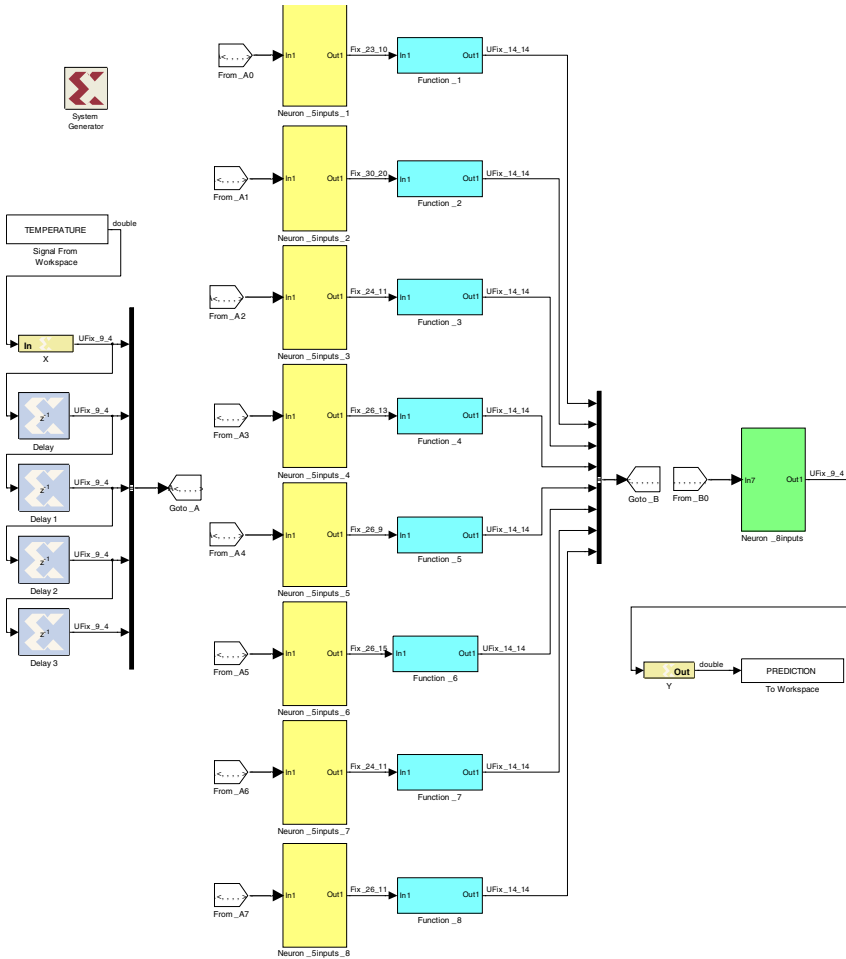


Fig. 4. Three layer Time Delay Neural Network architecture

Once the architecture has been defined, the values of the NN and its parameters are loaded with an initialization Matlab file. Initially the synaptic weights and bias are initiated with the values obtained in the training process. A maximum error of 1% is defined for the fixed point representation of these values. The input is analyzed to determine if the sign bit is necessary, the minimum number of bits of the integer part for covering the range and the minimum number of bits of the decimal part that keeps the error below the maximum permitted. For each synaptic weight and bias in the second layer the fixed point format is fixed in the same way. Same process is done for the final neuron. Therefore, logical resources are saved and optimal size of each neuron multiplier is reached because the weight and its input are optimized. Finally, in the initialization of the NN the unipolar sigmoid activation function is specified for all neurons in the second layer. The number of samples stored in the ROM is set and

the number of bits needed not to exceed the permissible error of 1%, so these values are unsigned 14 decimal bits.

Once the NN architecture has been defined and initialized System Generator simulations can be performed. The error in the temperature estimation is less than one degree centigrade. It is important to emphasize in this stage that the functionality of digital design has been checked.

The chosen device is a Spartan3E (part xc3s1200e, speed -5, package fg320), and the hardware description language is VHDL. The syntax of the VHDL files can be checked, and the synthesis and behavioral simulation of the system can be done with ISE software. After that, the implementation of the design allows the timing simulation of the NN (Fig. 5), the maximum combinational path delay is 74 ns. The ISE software provides a power estimator that indicates a power consumption of 0.36 watts in the FPGA, and an estimated case temperature of 27 degrees centigrade. The occupation rate of hardware in the FPGA is about 31.5% for logical resources and 7.6% for input-output pins. Lastly, the programming file is generated for the chosen FPGA.

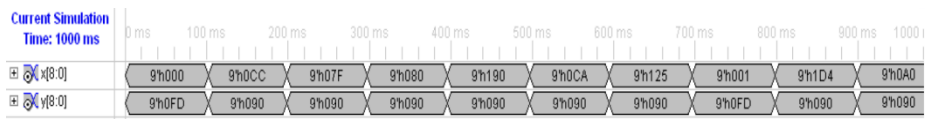


Fig. 5. Integrated System Environment timing simulation

4 Conclusions and Future Work

The reconfigurable prototype was developed on a FPGA and optimal physical performances were reached. The FPGA delay (74 ns) is much lower than time estimation with Matlab. The error in the FPGA temperature estimator is less than one degree centigrade. The design method uses bidirectional communication between Matlab and System Generator model. In the future the ANN parameters and the error allowed in the binary representation can be changed for checking the ANN performances. Prediction can be improved in the short and long term by increasing the sampling frequency and number of samples introduced in the TDNN.

Acknowledgements. This work has been supported by the Spanish Ministry of Foreign Affairs under the research project with reference D/033858/10.

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